Ideation Phase
Empathize & Discover

Date	31 January 2025
Team ID	LTVIP2026TMIDS39046
Project Name	Online Payments Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Empathy Map Canvas:

Empathy Map Canvas: An empathy map helps the team deeply understand users by capturing what they think, feel, see, hear, say, and do. It enables building solutions grounded in real user needs and constraints..

THINK & FEEL

- Is my online payment secure?
- What if someone misuses my card details?
- I hope the system detects fraud quickly.
- I don’t want to lose my money.
- Online transactions should be safe and fast.
- I trust digital payments but worry about fraud.

HEAR

- “There are many online payment scams.”
- “Use secure and trusted payment apps.”
- “Enable OTP verification.”
- “Banks monitor suspicious transactions.”
- News about increasing cyber fraud cases.

SEE

- OTP verification screens.
- Payment confirmation messages.
- Fraud alert notifications from bank.
- Transaction history in banking app.
- Security badges and encryption symbols.

SAY & DO

- Checks transaction details carefully.
- Verifies OTP before confirming payment.
- Contacts bank if suspicious activity occurs.
- Uses strong passwords.
- Enables two-factor authentication.

PAIN

- Fear of money loss.
- Delay in refund process.
- Unauthorized transactions.
- Complex complaint procedures.
- False fraud alerts blocking genuine payments.

GAIN

- Secure and safe transactions.
- Instant fraud detection alerts.
- Quick refund process.
- Trust in digital payment systems.
- Smooth and uninterrupted online payments.

How Our Solution Helps: The Machine Learning–based Online Payment Fraud Detection System analyzes transaction behavior patterns, detects anomalies in real-time, and flags suspicious activities instantly. This enhances transaction security, reduces financial loss, and builds trust for customers, banks, and payment platforms.